Beyond Multi-Stage Pipelines: A Label-Correlation Ensemble for Multi-Label Retinal Fundus Disease Classification

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***Abstract*—Retinal fundus images frequently carry more than one disease, and public fundus datasets are heterogeneous and severely long-tailed. Clinical triage suggests a staged design, in which a screening gate, a disease-group router and group-specific classifiers operate in sequence, but it is unclear whether this hierarchy earns its complexity. We present a controlled comparison of three architectures on a unified 39-target label space built from RFMiD 1.0, RFMiD 2.0 and ODIR-5K (12,652 images): (i) a three-stage gated cascade of eight models, (ii) a flat model with 39 independent sigmoid outputs, and (iii) a label-correlation model that replaces the flat linear head with 39 learned label queries and a two-layer transformer decoder. All three share one Swin-V2-B backbone, one preprocessing pipeline and one training protocol, so differences are attributable to the architecture. Staging offers no consistent advantage: tracing every missed disease instance to the stage that lost it shows that the two coarse gates discard 34.2% of true instances while the fine-grained classifiers lose only 6.7%, and the gates' hard zeros depress the cascade's macro-AUC to 0.747. Label-correlation modeling gives the largest single gain, raising the detection rate from 0.621 to 0.682 (0.689 with test-time augmentation). Blending the flat and label-correlation models under four-view test-time augmentation is the best configuration on six of eight metrics (exact match 0.694, micro-F1 0.678, macro-F1 0.456, macro-AUC 0.972). On definition-matched class subsets it matches or exceeds four recent studies, most clearly on low-support pathologies. We also show that ensembling a gated cascade with continuous-output models is harmful, and that per-source negative sampling is needed to remove a dataset shortcut that otherwise yields a spurious AUC of 1.000.**

***Index Terms*—Retinal fundus imaging, multi-label classification, hierarchical cascade, label correlation, transformer decoder, shortcut learning, test-time augmentation, ensembling.**

I. Introduction

Diabetic retinopathy, glaucoma and age-related macular degeneration are among the leading causes of preventable blindness; more than 2.2 billion people live with a vision impairment, nearly half of which could have been prevented or remains unaddressed [1]. Fundus photography is cheap and non-invasive, and deep networks reach specialist-level accuracy on single diseases such as diabetic retinopathy [2], [3]. Real screening, however, is a *multi-label* problem: one eye may show diabetic retinopathy, laser scars and media haze at once, and the clinically important tail contains dozens of conditions with only a handful of examples each [4], [5].

Two design philosophies compete for this problem. A *hierarchical cascade* mirrors clinical triage: a binary gate removes healthy eyes, a router assigns the remaining eyes to broad disease groups, and small group-specific classifiers make the fine-grained call. Each sub-problem is easier, and failures are attributable to a stage. A *flat* model predicts every target from one shared representation, avoiding early irreversible decisions. A third option keeps the flat topology but lets labels interact, so that evidence for one disease can inform another.

Prior multi-disease fundus studies usually compare across different backbones, preprocessing and label sets, which makes it impossible to separate the effect of the architecture from everything else. We hold everything except the architecture fixed and ask a direct question: *does the hierarchy earn its complexity?*

Contributions.

- A controlled comparison of a three-stage cascade, a flat classifier and a label-correlation transformer decoder on a unified 39-target label space merged from RFMiD 1.0, RFMiD 2.0 and ODIR-5K, with a shared backbone, preprocessing, loss, optimizer and calibration protocol.
- A *loss-cascade* analysis that attributes every missed disease instance to the stage that lost it, showing that the coarse gates, not the fine-grained classifiers, bound the cascade's recall, and that its hard-gated zeros invalidate rank-based metrics.
- A nine-configuration ablation isolating test-time augmentation (TTA), threshold re-tuning, label-correlation modeling and ensembling, including the negative result that a gated cascade is a harmful ensemble member.
- A source-balanced negative-sampling scheme that removes a provenance shortcut in the media-opacity group, and a subset-matched comparison with four recent studies.
- Cleaned code and the full test-set predictions of the best configuration, publicly released.

TABLE I
MERGED DATASET. EACH SOURCE KEEPS ITS OFFICIAL SPLIT.

Source	Train	Val	Test	Native labels
ODIR-5K (per eye) [7]	6,092	861	1,642	8 categories + keywords
RFMiD 1.0 [4]	1,920	640	640	45 conditions
RFMiD 2.0 [6]	508	177	172	50 conditions
Merged	8,520	1,678	2,454	39 scored targets
of which healthy	3,652	617	1,005	

II. RELATED WORK

Multi-disease fundus classification. RFMiD [4] and RFMiD 2.0 [6] annotate 45 and 50 conditions, and ODIR-5K [7] provides eight categories with per-eye diagnostic keywords. Cen *et al.* [5] detected 39 conditions with a hierarchical ensemble trained on a large private corpus. On public data, Rodríguez *et al.* [8] introduced the MuReD dataset and a C-Tran label-query transformer; Bhati *et al.* [9] targeted imbalance with discriminative kernel convolutions; Singh *et al.* [10] ensembled CNN and transformer backbones; and recent work benchmarks twelve CNN, transformer, hybrid and vision-language models on RFMiD [11], or combines dual CNN backbones [12] and asymmetric losses in a cascade [13]. None isolates the effect of hierarchy against a matched flat model.

Hierarchical classification exploits a label taxonomy by decomposing the task [14]; top-down cascades are known to propagate early errors, but the size of this effect in multi-label medical screening is rarely measured.

Label correlation. ML-GCN [15] propagates label embeddings over a co-occurrence graph, while C-Tran [16] and Query2Label [17] use learnable label queries in a transformer decoder. Our label-correlation model follows the query-decoder design.

Imbalance and shortcuts. Focal [18] and asymmetric [19] losses re-weight easy negatives. Models trained on pooled multi-site data can learn site or device signatures instead of pathology [20], [21], which motivates our per-source evaluation and negative sampling. TTA [22] and deep ensembles [23] are standard ways to stabilize predictions.

III. DATA AND LABEL SPACE

Harmonization. ODIR-5K is converted from patient-level records to per-eye images whose labels come from that eye's diagnostic keywords. The healthy flags of the three sources (RFMiD 1.0's inverted *Disease Risk*, RFMiD 2.0's *WNL*, ODIR's *N*) are unified into one *Normal* indicator. ODIR's "Other" category is removed because it is a catch-all rather than a definable pathology. Image identifiers are namespaced by source to prevent silent collisions, and the official splits are merged split-wise (Table I); the merged test split is sealed and read only for final evaluation.

Label space. The 38 named diseases with at least six training images are organized into six clinically motivated groups (Table II). The 21 ultra-rare classes with five or fewer training images are pooled into one G_{Rare} target that is detected but not sub-classified, giving 39 scored targets. The distribution is extremely long-tailed: diabetic retinopathy has 2,219 training images, while 16 of the 39 targets have fewer than ten test positives.

TABLE II
DISEASE GROUPS, TRAINING IMAGES PER GROUP AND VALIDATION PERFORMANCE OF THE STAGE-2 ROUTER (PER GROUP) AND THE STAGE-3 CLASSIFIERS (MACRO OVER THE GROUP).

			Stage 2		Stage 3	
Group	Diseases	Train	AUC	AP	AUC	AP
G1 Vascular	DR HR HTN BRVO CRVO TV CWS PRH MCA	2,554	.970	.969	.971	.673
G2 Exudative	ARMD DN EDN CSR CME RPEC MHL ERM PT	698	.932	.775	.926	.582
G3 Optic disc	ODC GLA ODP ODE AION TD ST	806	.902	.745	.942	.660
G4 Structural	MYA TSLN LS RT RD CF	694	.978	.933	.983	.919
G5 Media	MH CAT	656	.969	.874	.983†	.948†
G6 Inflamm.	CRS RS MS AH RP	162	.946	.587	.951	.858
G_{Rare}	21 pooled classes	42	.878	.177	–	–

†With source-balanced negatives; the positives-only model scores 1.000 / 1.000 through a source shortcut.

IV. METHODS

A. Problem formulation

Each image x has a target vector $\mathbf{y} \in \{0,1\}^{39}$; a healthy image has $\mathbf{y} = \mathbf{0}$, and an image may carry several diseases. Every system outputs scores $p_c(x) \in [0,1]$ and decisions $\hat{y}_c = \mathbf{1}[p_c(x) \geq \tau_c]$. For the cascade, let $s_1(x)$ be the Stage-1 probability of disease, $s_{2,g}(x)$ the router score of group g and $s_{3,c}(x)$ the Stage-3 score of disease $c \in g$. With hard gating,

$$p_c^{\text{casc}}(x) = \mathbf{1}[s_1(x) \geq \tau_1]\ \mathbf{1}\big[s_{2,g(c)}(x) \geq \tau_{g(c)}\big]\ s_{3,c}(x), \quad (1)$$

so any gate error forces $p_c = 0$ irrespective of the Stage-3 evidence. The flat and label-correlation models instead produce $p_c(x) = \sigma(f_c(x))$ for every c and every image. This difference drives both the recall ceiling of the cascade (Section V-A) and the behaviour of its rank-based metrics.

B. Shared protocol

All models use Swin-V2-B [24] with ImageNet-22k pre-training and ImageNet-1k fine-tuning, loaded through timm [25],[1] at 384×384. Preprocessing crops each image to the bounding box of the illuminated retina and zero-pads the short side to a square, so no retinal tissue is discarded. Training augmentation is random resized crop (scale 0.85–1.0), horizontal and vertical flips, $\pm 15°$ rotation and mild color jitter. Multi-label models use binary cross-entropy with a per-class positive weight $w_c = \min(\max(N_c^-/N_c^+, 1), 50)$. Optimization is

[1]Checkpoint `swinv2_base_window12to24_192to384.ms_in22k_ft_in1k`.

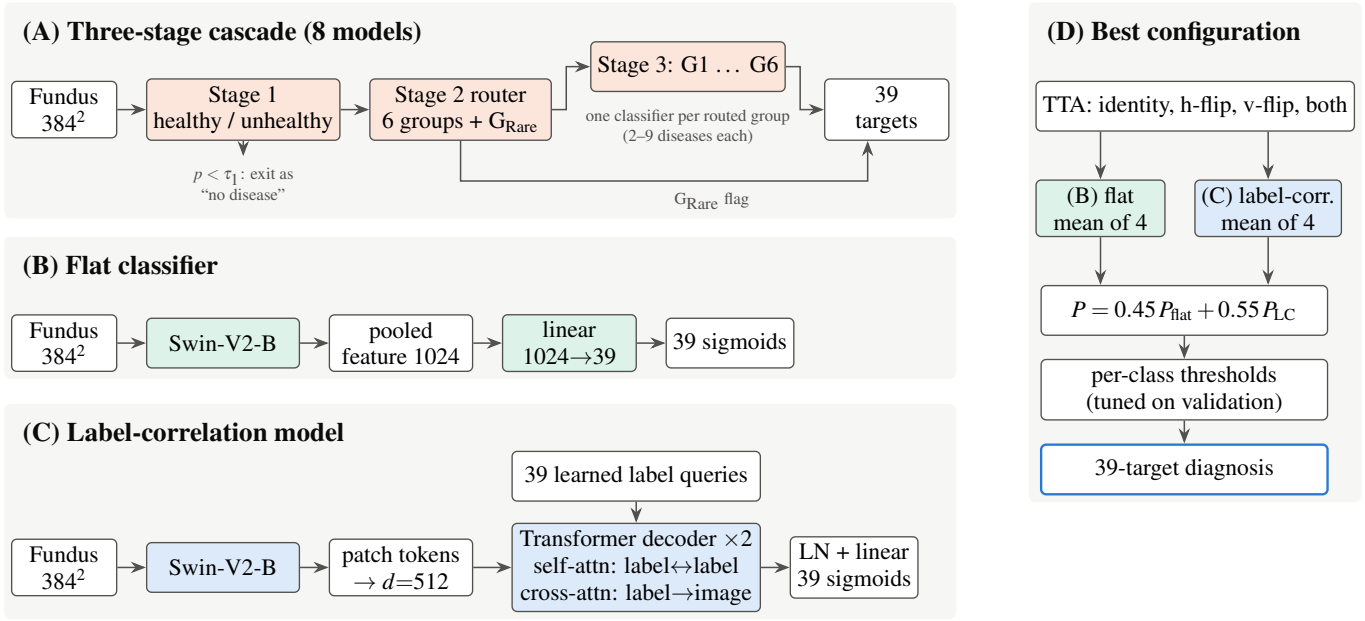


Fig. 1. The four systems compared. (A) Healthy images exit at Stage 1; unhealthy images are routed to every group that fires, and each routed group's classifier makes the fine-grained call. (B) One linear head predicts all 39 targets. (C) 39 label queries attend to the image tokens and to each other. (D) The best configuration blends (B) and (C) under four-view TTA.

TABLE III
TRAINING SETTINGS. ALL MODELS: SWIN-V2-B AT 384 PX, ADAMW, WEIGHT DECAY 10^{-2}, WARM-UP THEN COSINE DECAY, BATCH SIZE 4 WITH GRADIENT ACCUMULATION (BATCH: EFFECTIVE BATCH). [s] SOFTMAX; ALL OTHER HEADS ARE SIGMOID.

Model	Out.	Images	LR	Batch	Ep.	Pat.	Monitor
Stage 1	2^s	8,520	10^{-5}	8	60	8	AUC
Stage 2	7	4,868	$2{\cdot}10^{-5}$	8	30	7	macro-AUC
Stage 3 (×6)	2–9	162–2,554	$2{\cdot}10^{-5}$	8	30	7	macro-AP
Flat	39	8,520	$2{\cdot}10^{-5}$	8	30	7	macro-AP
Label-corr.	39	8,520	$2{\cdot}10^{-5}$	16	25	6	macro-AP

AdamW [26] (weight decay 10^{-2}) with linear warm-up and cosine decay, mixed precision, gradient clipping at 1.0, batch size 4 with gradient accumulation, and early stopping on the validation monitor. Decision thresholds are tuned per class on *validation* by maximizing F1 over $\{0.05, 0.075, \ldots, 0.95\}$ for every target with at least ten validation positives (22 of 39); the others use 0.5. The test split is used once. Table III lists the per-model settings.

C. Architecture A: three-stage cascade

Stage 1 is a two-way softmax screening gate (healthy vs. unhealthy) trained on all 8,520 training images with class-weighted cross-entropy (learning rate 10^{-5}, 60 epochs, patience 8; class weighting outperformed a weighted sampler, validation AUC 0.938 vs. 0.920). **Stage 2** is a multi-label router with seven sigmoid outputs (six groups and G_{Rare}) trained on the 4,868 unhealthy training images; an image is routed to every group whose score exceeds its tuned threshold. Among BCE, focal, asymmetric loss and GeM pooling, plain BCE gave the best validation macro-AP (0.722 vs. 0.703, 0.714, 0.710). **Stage 3** comprises six group-specific multi-label classifiers trained with ground-truth routing on the images of their group (2–9 classes each). G_{Rare} is decided at Stage 2 because its constituent classes are too scarce for a dedicated model. Gating is hard: an image rejected at Stage 1, or not routed to a group, receives probability zero for every downstream target, exactly as a deployed cascade behaves.

Source-balanced negatives. In G5, media haze comes almost entirely from RFMiD and cataract almost entirely from ODIR, so a positives-only G5 model reaches a validation AUC of 1.000 by recognizing the camera. We therefore add negatives (unhealthy images without a G5 disease) sampled *per source* in proportion to that source's positives, which equalizes the positive rate within every source. The de-confounded model scores a macro-AUC of 0.983; measured within single sources, where provenance cannot explain the result, the AUC is 0.987 for cataract in ODIR and 0.954 for media haze in RFMiD 1.0. This model is used in the cascade.

D. Architecture B: flat classifier

A single Swin-V2-B maps its 1,024-d pooled feature through a linear layer to 39 independent sigmoids. It is trained on all training images (healthy images have all-zero targets) and checkpointed on validation macro-AP.

E. Architecture C: label-correlation model

The flat linear head is replaced by a query decoder. Swin patch tokens are projected to $d = 512$ and serve as decoder memory. Thirty-nine learned query vectors, one per target, pass through a two-layer pre-norm transformer decoder (8

heads, feed-forward width 2,048, dropout 0.1); self-attention lets the labels exchange evidence and cross-attention lets each query inspect the image regions relevant to its disease. A LayerNorm and a shared linear layer map each query to one logit (95.8 M parameters vs. 86.9 M for the flat model). The training set contains 204 distinct co-occurring disease pairs (e.g., DR–HR, TSLN–MH, ODC–TSLN, DR–LS), which an independent sigmoid head cannot represent. Training is identical to the flat model except for a larger effective batch (16) and a shorter schedule (Table III). Both early-stopped after 19 epochs; the selected checkpoints have validation macro-AP 0.516 (flat) and 0.512 (label-correlation).

F. Architecture D: TTA and ensembling

TTA averages sigmoid outputs over four label-preserving views: identity, horizontal flip, vertical flip and both. Because averaging shifts the probability distribution, thresholds are re-tuned on validation TTA outputs. The ensemble is $P = w\,P_{\mathrm{LC}} + (1-w)\,P_{\mathrm{flat}}$, with w swept over $\{0, 0.05, \ldots, 1\}$ to maximize validation macro-AP; the selected $w = 0.55$ is applied unchanged to test, followed by per-class thresholds tuned on the blended validation outputs.

G. Metrics

We report exact-match (subset) accuracy, mean per-image Jaccard index, micro- and macro-F1, macro-F1 over the 23 well-supported targets ($\geq$ 10 test positives), macro-AUC, macro-AP and the *detection rate*, the fraction of all 1,719 true disease instances in the test split that are predicted positive. Macro averages run over the targets with at least one test positive. Every configuration is scored by the same routine on the same split.

V. Results

A. Where the cascade loses disease

Table V and Fig. 2 trace the cascade on the test split. Stage 1 reaches 0.852 accuracy and 0.928 AUC but sends 297 diseased images home as healthy, and every disease they carry is lost. Stage 2 is accurate on the images it receives (macro-AUC 0.955), yet a missed group makes all of its diseases unreachable. Of the 1,719 true disease instances, 34.2% are lost at the two coarse gates (18.6% at Stage 1, 15.6% at Stage 2) and only 6.7% by the fine-grained classifiers; 59.0% are detected. The coarse decisions, not the fine-grained ones, bound the cascade's recall. The Stage-1 threshold trades this loss against healthy recall (Table IV): moving it from 0.5 to 0.2 raises unhealthy recall from 0.845 to 0.876 at the cost of healthy recall (0.912 to 0.853), but even the most liberal setting misses about one diseased image in ten, and every miss removes all of that image's diseases.

Gated zeros invalidate rank metrics. The cascade's macro-AUC is 0.747 although its group and disease classifiers individually exceed 0.90 on validation. A true positive discarded at a gate receives a hard score of 0.000 and is ranked below every true negative with a small non-zero score, which breaks the probability ordering that AUC and AP measure.

TABLE IV
Stage-1 operating point on validation (1,061 unhealthy, 617 healthy images).

Threshold τ_1	.10	.20	.30	.50	.70	.90
Unhealthy recall	.896	.876	.862	.845	.825	.770
Healthy recall	.799	.853	.887	.912	.940	.968
Accuracy	.861	.867	.871	.870	.867	.843

TABLE V
Three-stage cascade on the test split ($n = 2{,}454$), all upstream errors propagated.

Stage 1 accuracy / AUC	.852 / .928
Stage 1 sensitivity / specificity	.795 / .933
Diseased images rejected at Stage 1	297 of 1,449
Stage 2 macro-AUC on received images	.955
Exact group set / exact disease set	.732 / .687
Images: correct healthy / exact set / partial	942 / 744 / 213
Images: total miss / false alarm	492 / 63
Instances lost at Stage 1 / 2 / 3	319 / 269 / 116
Instances detected	1,015 / 1,719 (59.0%)

Decision-based metrics (F1, exact match, detection) remain meaningful and are the appropriate basis for comparing a gated system.

Case studies. Table VI traces individual test images through both designs. In `4719_right`, Stage 1 assigns $s_1 = 0.243$ to an eye with diabetic retinopathy and the image exits as healthy; the flat model also misses it, so this is a hard image rather than a pure gating error. In `rfmid_1_20_test`, the router fires on the vascular and structural groups but not on the exudative group, so the macular hole becomes unreachable, the structural classifier finds the laser scars but not the tessellation, and the vascular classifier adds a false diabetic retinopathy; the flat model finds the laser scars without the false positive. Both designs flag glaucoma on the healthy eye `3903_left`, and both are exactly right on the multi-label image `rfmid_1_240_test`.

B. Does the hierarchy earn its complexity?

Table VII compares all nine configurations. Against the flat baseline (row 1), the cascade (row 4) wins slightly on the two image-level metrics, exact match (0.687 vs. 0.679) and Jaccard (0.728 vs. 0.723), because its screening gate cleanly passes healthy eyes. It loses on every instance-level metric: micro-F1 0.649 vs. 0.664, macro-F1 0.392 vs. 0.443 and detection 0.594 vs. 0.621, with the gap widest on the macro metrics that weight rare classes equally. With eight models to train, calibrate and maintain, the cascade buys interpretability of its failure modes, not accuracy.

C. What each technique contributes

Fig. 3 summarizes the ablation. **TTA** applied naively slightly reduces detection (-0.015, row 2) because the averaged probabilities no longer match thresholds tuned on single-view outputs; with thresholds re-tuned on validation (row 3)

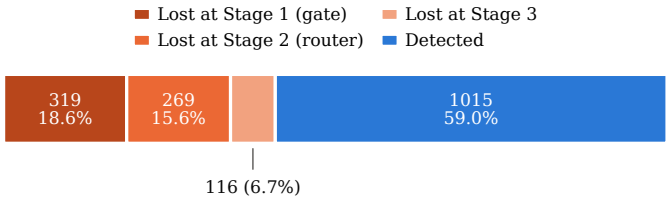


Fig. 2. Loss cascade: the stage at which each true disease instance of the test split is lost. The two coarse gates account for 34.2% of all instances.

TABLE VI
CASE STUDIES ON TEST IMAGES (✓ FOUND, × MISSED, FP FALSE POSITIVE).

Image	Truth	Cascade	Flat
4719_right	DR	exits at Stage 1 (s_1=0.24): DR ×	DR ×
rfmid_1_20	MHL, TSLN, LS	G2 not routed: MHL ×; TSLN × at Stage 3; LS ✓; DR FP	LS ✓; MHL ×; TSLN ×
3903_left	healthy	s_1=0.61, routed to G3: GLA FP	GLA FP
rfmid_1_240	TSLN, MH	TSLN ✓, MH ✓	TSLN ✓, MH ✓
2322_right	CAT	CAT ✓	CAT ✓

it improves detection by +0.029 and micro-F1 by +0.001. Re-tuning is a necessary part of TTA, not an option.

Label-correlation modeling is the largest single gain. With nothing changed but the head, detection rises from 0.621 to 0.682 (row 6) and to 0.689 with TTA (row 7, the highest of the study), and macro-AUC rises from 0.943 to 0.966. Detection had been the weakest metric of every flat configuration; co-occurrence modeling addresses it directly, consistent with its mechanism (evidence of diabetic retinopathy makes exudates and cotton-wool spots more findable).

Ensembling is additive only between continuous-output models. Blending the precision-oriented flat model with the recall-oriented label-correlation model (row 9) is best on six of eight metrics: exact match 0.694, Jaccard 0.740, micro-F1 0.678, macro-F1 0.456, well-supported macro-F1 0.594 and macro-AUC 0.972. In contrast, averaging the cascade with the flat model (row 5) is worse than the flat model alone on micro-F1 (0.654 vs. 0.665 for row 3) and detection, and the three-way ensemble (row 8) is worse still (micro-F1 0.646, detection 0.574). The cascade's sparse output matrix suppresses its partners' probabilities precisely on the diseased images its gates discarded. We report this as a negative result: gated pipelines are unsuitable members of probability-averaging ensembles.

D. Per-disease and per-source behaviour

Fig. 4 and Table VIII show that the best configuration leads or ties on most frequent targets; the cascade remains competitive on cataract (CAT), media haze (MH) and chorioretinitis (CRS), where its group specialists see a narrower problem. The best configuration exceeds an F1 of 0.80 on cataract (0.840),

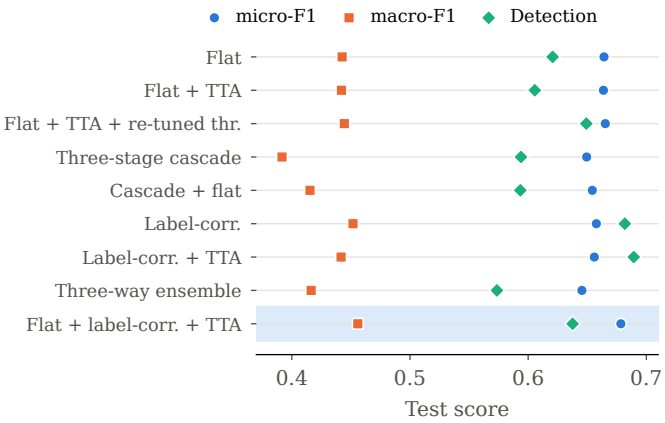


Fig. 3. Ablation at a glance (values of Table VII). The shaded row is the best configuration; the label-correlation rows carry the highest detection.

myopia (0.838) and laser scars (0.813) and reaches 0.785 on diabetic retinopathy, but hypertensive retinopathy (0.286) and macular scar (0.154) remain hard, and six ultra-low-support targets (PRH, AION, PT, ST, MCA, CF; ≤ 7 test positives) are never detected. Performance also depends on the source (Table IX): every model scores best on ODIR-5K, whose eight-category labels are coarse, and worst on RFMiD 2.0, whose 172 test images carry the richest and rarest label sets.

Two operating points. The label-correlation model with TTA is the high-sensitivity choice for population screening, where a missed disease costs more than a false alarm (detection 0.689). The flat + label-correlation ensemble is the balanced choice for decision support, with the best exact match and F1.

E. Comparison with recent studies

Our taxonomy matches no published class set, so we compare on *definition-matched* subsets: for each study we take the identically defined classes, recompute our metrics on exactly that subset from the released test-set predictions, and compare with the published figures (Table X). These comparisons are computed once on the sealed test split and are not used for model selection.

The best configuration matches or exceeds every comparator on macro-F1 and micro-F1 wherever they are reported; the margin is largest against the twelve-model benchmark [11] (+0.138 macro-F1, +0.193 micro-F1) and smallest against the asymmetric-loss cascade [13], with which macro-F1 and macro-AP are essentially tied. The class-level breakdown available for C-Tran (Table XI) shows where the advantage comes from: we lead on low-support classes such as optic disc pallor (ODP, +0.327 F1, where C-Tran detects none), laser scars (+0.286), chorioretinitis (+0.110) and CRVO (+0.096), while C-Tran leads on frequent, strongly co-occurring classes such as tessellation (-0.141), optic disc edema (-0.133) and BRVO (-0.117). Because the comparators use different test sets (MuReD's merged ARIA/STARE/RFMiD partition, or the 640-image RFMiD test set) and one is a non-peer-reviewed

TABLE VII

ABLATION ON THE TEST SPLIT (2,454 IMAGES, 39 TARGETS). ALL THRESHOLDS AND BLEND WEIGHTS ARE SELECTED ON VALIDATION. $F1_{WELL}$: MACRO-F1 OVER THE 23 TARGETS WITH ≥ 10 TEST POSITIVES. BEST VALUE PER COLUMN IN BOLD.

	Configuration	Exact	Jaccard	micro-F1	macro-F1	$F1_{well}$	macro-AUC	macro-AP	Detection
1	Flat baseline	.6789	.7229	.6642	.4426	.5767	.9430	.5048	.6207
2	Flat + TTA×4	.6817	.7248	.6637	.4420	.5707	.9470	.5144	.6056
3	Flat + TTA×4 + re-tuned thresholds	.6764	.7273	.6653	.4444	.5748	.9470	.5144	.6492
4	Three-stage cascade	.6870	.7278	.6495	.3917	.5248	.7467*	.3713*	.5939
5	Cascade + flat ensemble ($w = 0.5$)	.6817	.7248	.6543	.4154	.5615	.9500	.5213	.5934
6	Label-correlation	.6679	.7239	.6577	.4518	.5573	.9662	.4599	.6818
7	Label-correlation + TTA×4	.6756	.7302	.6560	.4417	.5568	.9674	.4834	**.6894**
8	Three-way ensemble (equal weights)	.6838	.7229	.6455	.4164	.5220	.9710	**.5307**	.5736
9	Flat + label-corr. ensemble + TTA ($w_{LC} = 0.55$)	**.6944**	**.7399**	**.6784**	**.4559**	**.5942**	**.9716**	.5222	.6376

*Depressed by hard-gated zero probabilities. Row 4 scores the cascade's probabilities at 0.5; its native per-class decisions give micro-F1 .6451, macro-F1 .3922 and detection .5905.

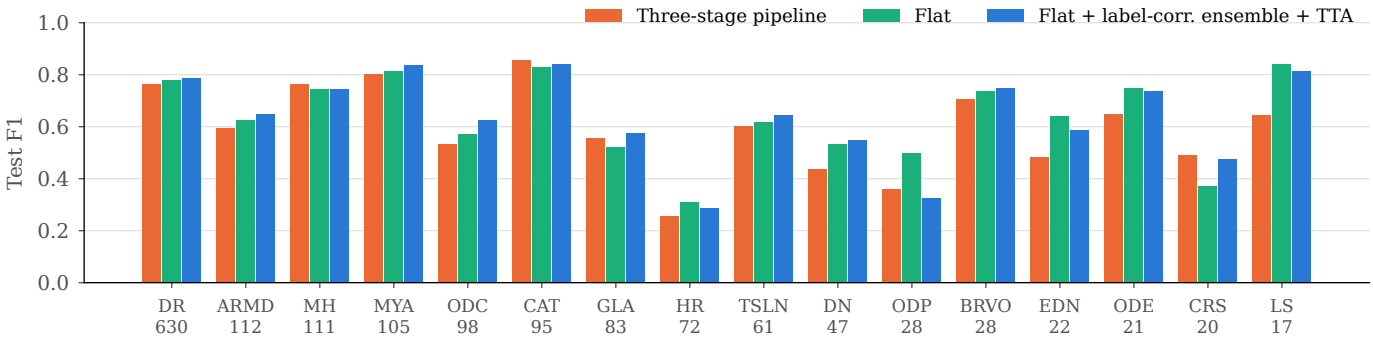


Fig. 4. Test F1 for the 16 most frequent targets (number of test positives under each code).

preprint [13], these results situate our model rather than rank it. Our macro-AUC (0.972 over all 39 targets) cannot be restricted to the matched subsets and is not compared.

VI. DISCUSSION AND LIMITATIONS

Staged versus end-to-end design. The cascade and the flat model are numerically close overall but behave very differently. The cascade decouples the 39-target problem into local decisions: each Stage-3 classifier searches a space of 2–9 classes, the router follows a clinical taxonomy, and every error can be localized to a gate, which is valuable for auditing. Its stages can be retrained or replaced independently. The price is eight networks to train, calibrate and maintain, and above all the sequential gating loss of Fig. 2: an instance discarded upstream is unrecoverable. End-to-end models score every image against every target with one or two networks, which removes gating losses and simplifies calibration and deployment, at the cost of intermediate interpretability and of retraining the whole model for any change.

Why the ensemble works. The two members have opposite biases. The label-correlation model is recall-oriented: it fires on learned co-occurrence and reaches the highest detection of the study, occasionally producing false positives when a secondary condition resembles a primary one. The flat model evaluates each target independently and is more conservative. With weight 0.55 on the label-correlation model, the blend keeps most of its recall gain on rare and co-occurring diseases, while the flat component suppresses spurious activations, which explains why exact match and every F1 measure improve together.

Shortcuts in pooled data. Merging datasets enlarges the training set but couples labels to provenance. The perfect G5 score of a positives-only model is a pure artifact, and it is invisible in pooled validation metrics. Reporting within-source metrics and sampling negatives per source should be standard practice whenever a label is concentrated in one source.

Limitations. The ultra-rare classes remain unsolved across all architectures and losses, which points to data scarcity, not model capacity, as the binding constraint. Sixteen of the 39 targets have fewer than ten test positives, so their per-class figures are statistically unstable. The merged test split is dominated by ODIR-5K, on which all models score best, so aggregate figures are not directly comparable with single-dataset studies. The intermediate disease groups follow the lesion literature but were not reviewed by an ophthalmologist. The cascade's components received more ablation (losses, pooling, samplers) than the flat and label-correlation models, which were each trained once, an asymmetry that favors the cascade. Ensemble weights and thresholds are tuned on the same validation split that selects checkpoints, so a small optimistic bias cannot be excluded. Future work should replace hard gates with soft probabilistic routing, bring label-correlation decoding into the router, and address the rare tail with multi-center data or

TABLE VIII

PER-TARGET TEST RESULTS, ORDERED BY NUMBER OF TEST POSITIVES (n). F1 OF THE THREE-STAGE CASCADE (NATIVE DECISIONS), THE FLAT MODEL AND THE BEST CONFIGURATION (FLAT + LABEL-CORRELATION ENSEMBLE + TTA); AP AND DECISION THRESHOLD OF THE BEST CONFIGURATION ("–": DEFAULT 0.5, FEWER THAN TEN VALIDATION POSITIVES). BEST F1 PER TARGET IN BOLD.

Target	n	Cascade	Flat	Best	AP	Thr.	Target	n	Cascade	Flat	Best	AP	Thr.
DR	630	.762	.778	**.785**	.880	.70	RT	13	.316	.375	**.421**	.367	.35
ARMD	112	.594	.627	**.650**	.657	.40	CRVO	11	.632	**.667**	.667	.669	.72
MH	111	**.764**	.745	.745	.854	.90	CME	10	.286	.444	**.588**	.636	–
MYA	105	.802	.812	**.838**	.902	.68	TV	8	.000	.182	**.235**	.276	–
ODC	98	.532	.571	**.627**	.687	.80	CWS	8	.154	.400	**.500**	.363	–
CAT	95	**.856**	.829	.840	.922	.55	TD	8	**.375**	.182	.154	.211	–
GLA	83	.555	.522	**.573**	.584	.95	PRH	7	**.250**	.000	.000	.300	–
HR	72	.255	**.309**	.286	.286	.37	AH	6	.286	.444	**.571**	.604	–
TSLN	61	.602	.619	**.646**	.684	.85	ERM	6	.000	.222	**.400**	.353	–
DN	47	.436	.532	**.549**	.555	.85	AION	5	**.222**	.000	.000	.067	–
ODP	28	.361	**.500**	.327	.386	.55	RPEC	5	.000	.000	**.167**	.155	–
BRVO	28	.706	.737	**.750**	.797	.88	MHL	4	.190	**.400**	.286	.291	–
EDN	22	.481	**.640**	.588	.700	.12	PT	4	**.400**	.000	.000	.126	–
ODE	21	.647	**.750**	.737	.760	.62	RP	3	.000	**.500**	**.500**	.867	–
CRS	20	**.489**	.370	.474	.540	.57	ST	3	.000	.000	.000	.016	–
LS	17	.643	**.839**	.812	.813	.57	HTN	2	.000	**.667**	.500	.583	–
CSR	16	.444	**.562**	.552	.632	.65	MCA	2	**.667**	.000	.000	.550	–
G_{Rare}	15	.261	**.364**	.364	.336	.72	CF	2	.000	.000	.000	.044	–
RS	15	.667	.560	**.692**	.743	.72	RD	2	.571	**1.00**	.800	1.00	–
MS	14	.091	.111	**.154**	.173	.55							

TABLE IX

TEST PERFORMANCE BY SOURCE DATASET.

		Flat		Three-stage cascade		
Source	n	macro-F1	Exact	macro-F1	Exact	Detection
ODIR-5K	1,642	.625	.746	.614	.757	.611
RFMiD 1.0	640	.375	.566	.335	.567	.602
RFMiD 2.0	172	.319	.459	.272	.465	.447

TABLE X

SUBSET-MATCHED COMPARISON (OURS / THEIRS). n: SHARED CLASSES.

Study	Data	n	macro-F1	micro-F1	macro-AP
C-Tran [8]	MuReD	15	**.657** / .638	–	**.704** / .685[a]
Dual CNN [12]	MuReD	15	**.657** / .403	**.726** / .548	**.704** / .582[b]
ASL cascade [13]	RFMiD	26	**.478** / .476	**.702** / .626	.530 / **.539**
12 models [11]	RFMiD	26	**.478** / .340	**.702** / .509	–

[a]Averaged over the comparator's full label set. [b]Micro PR-AUC. For [11], the best of twelve models per metric.

TABLE XI

PER-CLASS F1 AGAINST C-TRAN [8] ON THE 15 SHARED CLASSES (n: OUR TEST POSITIVES).

Class	n	Ours	C-Tran	ΔF1
DR	630	.785	.859	−0.074
MH	111	.745	.724	+0.021
ODC	98	.627	.703	−0.076
TSLN	61	.646	.787	−0.141
ARMD	112	.650	.615	+0.035
DN	47	.549	.607	−0.058
MYA	105	.838	.872	−0.034
BRVO	28	.750	.867	−0.117
ODP	28	.327	.000	+0.327
CRVO	11	.667	.571	+0.096
RS	15	.692	.706	−0.014
ODE	21	.737	.870	−0.133
LS	17	.812	.526	+0.286
CSR	16	.552	.500	+0.052
CRS	20	.474	.364	+0.110

generative augmentation.

VII. CONCLUSION

Under a controlled protocol on 39 retinal targets, a three-stage cascade offers no consistent advantage over a flat classifier: its coarse gates lose 34.2% of true disease instances, five times more than its fine-grained classifiers, and its hard zeros make rank-based metrics invalid. Replacing the flat head with a label-query transformer decoder gives the largest single improvement, and blending it with the flat model under four-view TTA yields the best overall system (micro-F1 0.678, macro-F1 0.456, macro-AUC 0.972). Gated pipelines should not be ensembled with continuous-output models, and pooled multi-source data must be checked within each source for provenance shortcuts.

DATA AND CODE AVAILABILITY

All datasets are public (RFMiD 1.0, RFMiD 2.0, ODIR-5K) and de-identified by their providers; no new data were collected. Code, the harmonization rules and the test-set predictions of the best configuration are available at https://github.com/asifuddin01/Beyond-Multi-Stage-Pipelines. The system is a research prototype, not a medical device.

REFERENCES

[1] World Health Organization, *World Report on Vision*. Geneva, Switzerland: WHO, 2019.

[2] V. Gulshan *et al.*, "Development and validation of a deep learning algorithm for detection of diabetic retinopathy in retinal fundus photographs," *JAMA*, vol. 316, no. 22, pp. 2402–2410, 2016.

[3] D. S. W. Ting *et al.*, "Development and validation of a deep learning system for diabetic retinopathy and related eye diseases using retinal images from multiethnic populations with diabetes," *JAMA*, vol. 318, no. 22, pp. 2211–2223, 2017.

[4] S. Pachade *et al.*, "Retinal fundus multi-disease image dataset (RFMiD): A dataset for multi-disease detection research," *Data*, vol. 6, no. 2, p. 14, 2021, doi: 10.3390/data6020014.

[5] L.-P. Cen *et al.*, "Automatic detection of 39 fundus diseases and conditions in retinal photographs using deep neural networks," *Nature Commun.*, vol. 12, Art. no. 4828, 2021, doi: 10.1038/s41467-021-25138-w.

[6] S. Panchal *et al.*, "Retinal fundus multi-disease image dataset (RFMiD) 2.0: A dataset of frequently and rarely identified diseases," *Data*, vol. 8, no. 2, p. 29, 2023, doi: 10.3390/data8020029.

[7] Peking University, "Ocular disease intelligent recognition (ODIR-5K)," 2019. [Online]. Available: https://www.kaggle.com/datasets/andrewmvd/ocular-disease-recognition-odir5k

[8] M. A. Rodríguez, H. AlMarzouqi, and P. Liatsis, "Multi-label retinal disease classification using transformers," *IEEE J. Biomed. Health Inform.*, 2022, doi: 10.1109/JBHI.2022.3214086.

[9] A. Bhati, N. Gour, P. Khanna, and A. Ojha, "Discriminative kernel convolution network for multi-label ophthalmic disease detection on imbalanced fundus image dataset," arXiv:2207.07918, 2022.

[10] D. Singh, S. Agarwal, and S. Mishra, "Retinal fundus multi-disease image classification using hybrid CNN-transformer-ensemble architectures," arXiv:2503.21465, 2025.

[11] D. Dey, A. Ajbar, and Y. Yan, "Benchmarking convolutional, transformer, hybrid, and vision language models for multi disease retinal screening," in *Proc. Int. Conf. Medical and Health Informatics (ICMHI)*, 2026, arXiv:2605.26283.

[12] M. A. Mahmood *et al.*, "Integrated deep learning model for multi-label retinal disease diagnosis," *Sci. Rep.*, vol. 16, Art. no. 21847, 2026, doi: 10.1038/s41598-026-54982-3.

[13] A. Saikia *et al.*, "A unified deep learning cascade for retinal disease detection and diabetic retinopathy severity grading," Research Square preprint, 2026, doi: 10.21203/rs.3.rs-9980883/v1.

[14] C. N. Silla and A. A. Freitas, "A survey of hierarchical classification across different application domains," *Data Mining Knowl. Discovery*, vol. 22, no. 1–2, pp. 31–72, 2011.

[15] Z.-M. Chen, X.-S. Wei, P. Wang, and Y. Guo, "Multi-label image recognition with graph convolutional networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2019, pp. 5177–5186.

[16] J. Lanchantin, T. Wang, V. Ordonez, and Y. Qi, "General multi-label image classification with transformers," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2021, pp. 16478–16488.

[17] S. Liu, L. Zhang, X. Yang, H. Su, and J. Zhu, "Query2Label: A simple transformer way to multi-label classification," arXiv:2107.10834, 2021.

[18] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, 2017, pp. 2980–2988.

[19] T. Ridnik *et al.*, "Asymmetric loss for multi-label classification," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2021, pp. 82–91.

[20] J. R. Zech *et al.*, "Variable generalization performance of a deep learning model to detect pneumonia in chest radiographs: A cross-sectional study," *PLoS Med.*, vol. 15, no. 11, Art. no. e1002683, 2018.

[21] R. Geirhos *et al.*, "Shortcut learning in deep neural networks," *Nature Mach. Intell.*, vol. 2, pp. 665–673, 2020.

[22] D. Shanmugam, D. Blalock, G. Balakrishnan, and J. Guttag, "Better aggregation in test-time augmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2021, pp. 1214–1223.

[23] B. Lakshminarayanan, A. Pritzel, and C. Blundell, "Simple and scalable predictive uncertainty estimation using deep ensembles," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2017.

[24] Z. Liu *et al.*, "Swin Transformer V2: Scaling up capacity and resolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2022, pp. 12009–12019.

[25] R. Wightman, "PyTorch image models," 2019. [Online]. Available: https://github.com/huggingface/pytorch-image-models

[26] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2019.